

L'ORÉAL – Anti-Diversion Code (AD Code) Management

System: Speed WMS

Module: Multi-Order Preparation (SPD011)

Speed Version: A.8.6.F (28.06b1)

Radio Version: 28.06b

1. Document Control

1.1 Version History

Version	Description	Date	Author
1.0	Initial Version	07/04/2025	AKD
1.1	Update after Rémi email feedback	09/04/2025	AKD

2. Business Context

Within the L'Oréal activity, shipped cartons are traced using an Anti-Diversion Code (AD Code).

This is:

- A unique identifier
- Generated during preparation
- Used to track stock origin

The use of AD Codes is mandatory based on parameters defined in this SPD.

3. Activation Rules

The development is applied only if:

ACT_TOP13 = 1

AND

ART_TOP1 = 1 (AD Code management activated at article level)

There is no AD management in Speed Admin.

4. AD Code Level Determination

The AD Code requirement depends on:

- Picking type (MIL_TYPR)
- Article family (GRP_NIV1 / GRP_NIV2)
- Activity parameters (ACT_TOP1 / ACT_TOP2)

5. Picking Types (MIL_TYPR)

Code	Meaning	Level
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002	Unit Picking	SKU
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006	Full Pallet	PALLET
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001	Stock Picking	SKU or CARTON (depending on TOP parameters)
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6. Family Logic

AD Codes apply only if:

GRP_NIV1 = DIVERSION

And:

GRP_NIV2 = 10 (LPD)

GRP_NIV2 = 30 (PPD)

7. Management Level Rules

7.1 Type 006 – Full Pallet

- Always PALLET level
 - One AD Code per pallet
 - One AD Code per support
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7.2 Type 002 – Unit Picking

If GRP_NIV2 = 10 (LPD)

→ One AD Code per unit

Example:

15 units picked

→ 15 AD Codes required

Each AD Code:

- Must be unique in SPE_AD
 - BP123 allowed (bypass code)
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If GRP_NIV2 = 30 (PPD)

→ One AD Code per CARTON

Steps:

1. Calculate total units picked
2. Calculate number of cartons:

Number of cartons = Total Units / ART_QTEC

(Round up)

Example:

15 units

7 units per carton

$15 / 7 = 2.14 \rightarrow 3$ cartons

→ 3 AD Codes required

Each AD Code:

- Unique within current preparation
 - Not necessarily unique in SPE_AD (if incomplete cartons)
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7.3 Type 001 – Stock Picking

Family LPD (GRP_NIV2=10)

Controlled by:

ACT_TOP1

ACT_TOP1 Management Level

1 CARTON

0 or NULL SKU

Family PPD (GRP_NIV2=30)

Always CARTON management.

8. AD Code Validation Rules

When validating mission line:

Call:

PE30

After mission line validation.

If conditions met:

- ACT_TOP13=1
- ART_TOP1=1
- GRP_NIV1=DIVERSION

Then display new AD Code screen.

9. AD Code Scanning Screen Logic

- Based on serial number screen
- Adapted for AD Code

Behavior:

- Each scan adds AD Code to list
- Scanning existing code removes it

- Empty scan validates screen
- Cannot proceed if required quantity not reached

Error message:

"Veuillez flasher tous les AD Codes"

"Please scan all the AD Codes"

10. AD Code Format Configuration

Activity parameters:

Parameter	Description
ACT_ALPHA3	Prefix SKU
ACT_ALPHA4	Prefix CARTON
ACT_ALPHA5	Prefix PALLET
ACT_ALPHA6	EAN128 SKU Code (91–99)
ACT_ALPHA7	EAN128 CARTON Code
ACT_ALPHA8	EAN128 PALLET Code
ACT_NUM1	Length SKU
ACT_NUM2	Length CARTON
ACT_NUM3	Length PALLET

Priority Rule

If EAN128 parameter is configured (90–99):

→ EAN128 decoding overrides prefix/length validation.

Else:

Validate:

- Length matches ACT_NUMx

- Prefix matches ACT_ALPHAx

Only if ACT_NUMx and/or ACT_ALPHAx are defined.

11. Bypass Code

If AD Code cannot be read:

Operator can scan:

BP123

Rules:

- Stored like normal AD Code
 - Counts as valid
 - Ignores activity format rules
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12. AD Code Storage – Table SPE_AD

New specific table:

SPE_AD

Field	Description
ACT_CODE	Activity Code
MIL_KEYU	Mission Line KEYU
MIE_NOMI	Mission Number
MIL_NOLM	Mission Line Number
OPE_REDO	Client Code
OPE_DAT.TIE_NOM	Client Name
QUANTITY	Quantity linked to AD Code
ART_CODE	Article Code
AD_CODE	Scanned AD Code

Field	Description
AD_VL	SKU / CARTON / PALLET
AD_STAT	040
AD_JOUR	Date (YYYYMMDD)

12.1 Quantity Allocation Rule (Carton Example)

Example:

Carton size (ART_QTEC) = 10 units

Total picked = 33 units

Allocation:

AD Code Quantity

AD1 10

AD2 10

AD3 10

AD4 3

Remaining quantity assigned to last AD Code.

13. PE30 – Technical Logic Summary

IF ACT_TOP13 = 1

 IF ART_TOP1 = 1

 IF GRP_NIV1 = DIVERSION

 IF GRP_NIV2 = 10 (LPD)

 Apply LPD rules

 ELSE IF GRP_NIV2 = 30 (PPD)

 Apply PPD rules

Inside each:

IF MIL_TYPR = 006 → PALLET logic

IF MIL_TYPR = 002 → SKU or CARTON logic

IF MIL_TYPR = 001 → Based on ACT_TOP1

Block validation if required AD Code quantity not met.

14. Key Uniqueness Rules

Level	Uniqueness Requirement
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SKU	Unique in SPE_AD
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PALLET	Unique in SPE_AD
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CARTON	Unique per preparation, reusable across preps (if incomplete cartons)
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15. Client Validation

L'Oréal acknowledges and validates the content of this document.

BK Systèmes commits to delivering the solution as described.

Any additional request outside this scope may require a separate quotation.

Signed at: Vaulx-Milieu

Date: 07/04/2025

Company Signature & Stamp